Experiment-4 Straight Line Motion with a Constant Acceleration

In this part of the experiment, you will examine straight-line motion of an object (puck) with constant acceleration on an inclined frictionless air table. By plotting the experimental data, you will find the acceleration of the puck sliding down on an inclined air table.

Theory

When a particle slides straight down a frictionless inclined plane, its acceleration is constant, and it will move in a straight line down the plane. The magnitude of the acceleration depends on the angle at which the plane is inclined. If the inclination angle is 90^0, the object will slide down with an acceleration which is equal to the Earth's **gravitational acceleration** g with the magnitude of 9.8 m/s^2.

If you put the puck at the top of the inclined air table and let it slide down the plane, it will move downwards on a straight line but with increasing velocity. The rate of change of the velocity is the acceleration of the puck.

If at time t_1 the puck is at the point x_1 with a velocity of v_1, then at a later time t_2, it will be at a point of x_2 with a velocity v_2. For the motion, the average acceleration of the puck in this time interval $\Delta t = t_2 - t_1$ is defined as:

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1} \qquad (4)$$

*To investigate the one-dimensional motion of a puck on an air table, we choose a point (dot) near the **beginning** of the motion as the origin of the coordinate system. We determine the location of the puck moving in one dimension by the position x of sparks on the recording paper at **a constant time interval** between the marks on the data sheet.*

Just as **velocity** describes the rate of change of position with time, **acceleration** describes the rate of change of velocity with time. This means that the puck's velocity increases as it goes down the incline.

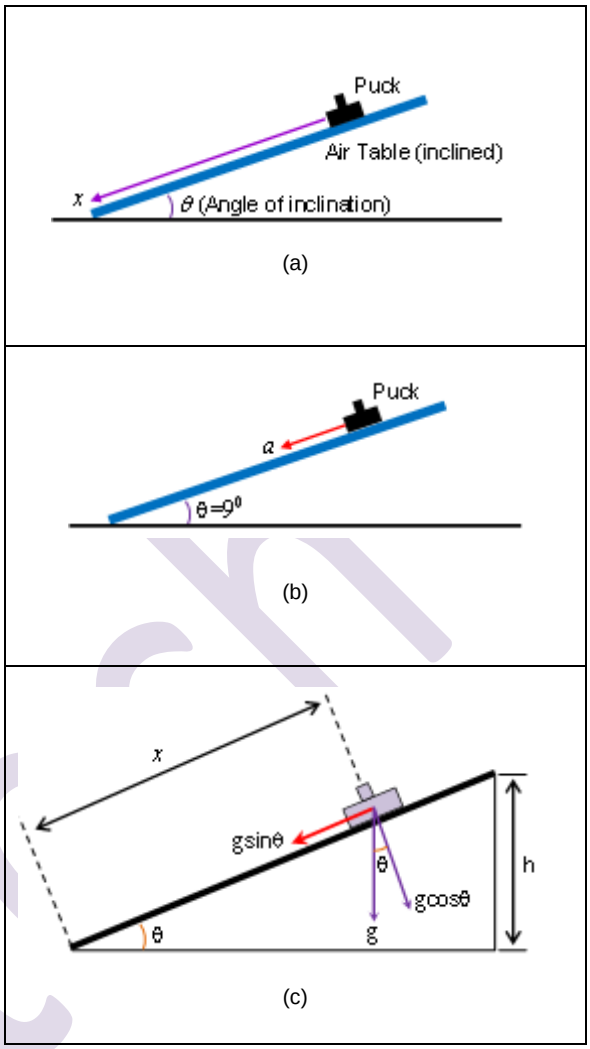


Figure-6: The set-up for a puck moving down an inclined air table (**a**), the straight down motion on an inclined air table (**b**) and components of **g**, the acceleration due to gravity **(c)**. Since the table is inclined at an angle **θ**, the puck's acceleration along the incline should be ***a*=gsinθ**.

If we consider the instantaneous acceleration (simply the acceleration) of the motion in the x-direction, it would be:

$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} \qquad (5)$$

In words, the acceleration of an object at any instant is the rate at which its velocity is changing at that instant.

In this experiment, we will observe the motion of a puck moving in a straight line with a velocity changing uniformly. The air table is inclined at an angle of θ with the horizontal plane as shown in the **Figure-(6)**.

The instantaneous acceleration in a straight line motion equals the instantaneous rate of **change of velocity** with time. Since velocity is a vector, acceleration is a vector too. But for one-dimensional motion, we need only use a plus or minus sign to indicate acceleration **direction** relative to a chosen coordinate axis. For the inclined air table, the positive x-axis is taken to be in the **direction** of the puck's motion.

Now, we examine the situation when the acceleration is constant and the motion is in a straight line. In this case, the instantaneous and average accelerations **are equal**. Suppose that at an initial time t_1=0, the puck is at the position of x_0 and has a velocity of v_1=v_0. At a later time t_2=t, it is in position x and has a velocity of v_2=v. If the acceleration of the puck is constant, the **average** acceleration and the **instantaneous** acceleration are **equal** to each other and therefore we find:

$$a = \frac{v - v_0}{t - 0} \qquad (6)$$

Then, the **velocity** of the puck will be:

$$v = v_0 + at \qquad (7)$$

Here, v_0 is the velocity at time t=0 and v is the velocity at any later time t.

For the interval from **t=0** to the later time **t**, the equitation for an object's **motion** with constant acceleration in one dimension can be written as:

$$x = x_0 + v_0 t + \frac{1}{2} a t^2 \qquad (8)$$

where:

x_0 : The displacement at time **t=0** (initial position),

v_0 : The velocity at time **t=0** (initial speed) and,

a : The object's acceleration.

In this experiment, we will allow a puck to accelerate down the incline while the sparktimer is in operation, but stop the sparktimer as the puck hits the bottom part of the inclined plane.

Once a spark (dot) has been selected on the recording paper, we choose a point near the beginning of the track as the **origin** and label it as **x=0** corresponding time **t=0**. Since the motion of object (puck) starts from rest (x_0=0, v_0=0) at **t=0**, the object's position at any time t (instantaneous position) will be:

$$x = \frac{1}{2} a t^2 \qquad \textbf{(Experimental)} \qquad (9)$$

Note that the *positive* x-axis is taken to be in the direction of the puck's motion. Using the displacement (x) from the *origin* to each data point, for each dot; the coordinate x should be measured from the origin to that point.

A graph of Equation-(9), that is, a x-t graph for motion with constant acceleration, is always a parabola that passes through the origin in the x-y plane. However, If a graph of x-t^2 is plotted, we find a straight line which has a **slope** of:

- "$\frac{1}{2}a$"

and it will pass through the origin.

As seen in the Figure-(6c), the acceleration (a) of an object (puck) moving down an inclined plane is a component of the acceleration due to gravity (g). The component of g pointing parallel to the plane causing the acceleration a is:

- $a = g \sin\theta$

- ✓ **So, if θ is known and acceleration (a) is determined, the acceleration due to gravity (g) can be calculated experimentally.**
- ✓ **If θ and g are assumed known, the expected value for the acceleration (a) can be determined.**

4.1. Experimental Procedures

1. Place a sheet of carbon paper and a sheet of recording paper (data sheet) on the air table with the recording paper on top.

2. First, level off the air table and then tilt it to an inclined position by using the block to give an inclination angle (the angle with the horizontal plane) as:
   - $\theta = 9^0$

3. Keep one of the pucks at rest over the carbon paper at the lower right corner of the air table. In this experiment, only one puck will be used.

4. Turn on the compressor (air supply).

   4.1. Put the puck at the top of the inclined plane of the air table and press the compressor pedal (switch).
   4.2. Check if the puck is falling **freely** down the plane.

5. Now, turn on the spark timer and set the spark timer frequency to:
   - f=20Hz.

> Do **not** touch the conducting portions, high voltage pins or the hoses of the pucks and the recording or the carbon paper **while spark timer is working**.

6. Position a puck at the raised end of the inclined air table, close to the middle.

   6.1. Press both puck and spark timer switches simultaneously, release the puck and **stop** pressing when the puck reaches the bottom part of the inclined plane. **Release the puck with no initial velocity**.
   6.2. **Turn off the spark-timer and then remove the puck from the air table.**
   6.3. Remove the data recording paper and examine the dots produced on it.
   6.4. The type of motion that you have observed is the straight line motion with constant acceleration (***acceleration does not change by the time***).

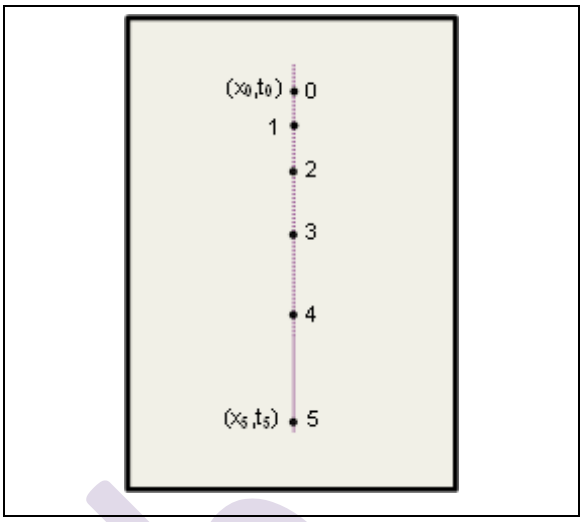


Figure-7: The dots produced by the puck on the data sheet for an inclined air table.

   6.5. Take the trajectory of the puck as the **positive** x-axis (Figure-7).
   6.6. Number and circle the dots (data points**) starting from the first dot** as **0,1,2,…..,10**. Take x**=0** and **t=0** at the first dot as your initial data point.
   6.7. Now, using a ruler, measure the **distance** (**x**) of the remaining dots from the **dot 0**.
   6.8. Also determine and record the **time** (**t**) for each of these dots.
   6.9. Record the data values in the data Table-(4) with respect to your initial point.
   6.10. Calculate and record t^2 values in the Table-(4).

7. **Plot** x versus t^2. Then, draw the best line that fits your data points.

8. Using the slope of this line, determine the **experimental** acceleration (a) of the puck. Show these calculations on your graph paper clearly.

   Note that:
   - $x = \frac{1}{2}at^2$ if $x_0 = 0$.
   - Then, the **slope** of the x versus t^2 graph will be equal to: "$\frac{1}{2}a$".

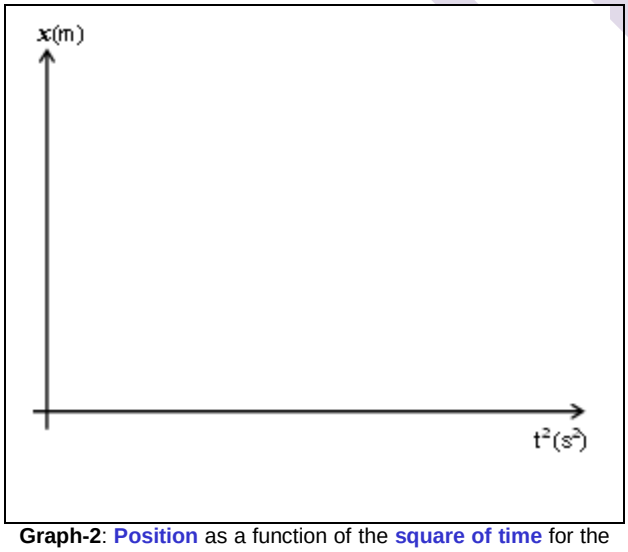
9. Compare your determination of the experimental acceleration (a) with the **expected** value, **gsinθ** where **g**=9.80m/s^2.

4.2. LABORATORY REPORT

Straight Line Motion with a Constant Acceleration

Table-4: Experimental data values for the straight line motion with a **constant acceleration**.

Inclination Angle θ (deg)	Frequency f (Hz)	Dot Number	$x(m)$	$t(\text{sec})$	$t^2(s^2)$	$a(m/s^2)$
			(Measured)			(By Slope)
…..	…..	0	$x_0 = 0$	$t_0 = 0$	0	…..
		1				
		2				
		3				
		4				
		5				
		6				
		7				
		8				
		9				
		10				



Graph-2: **Position** as a function of the **square of time** for the straight line motion with a constant acceleration.

Conclusions and Questions

- Determine the uncertainty in your measurement of displacements between dots on the data sheet?.
- By the slope, what is the **experimental** value of the acceleration (a) for the motion, in units of **m/s^2?**. Is the acceleration of the motion constant?.
- Write the expression for the **expected** acceleration of a particle down a smooth plane, inclined at an angle θ with respect to the horizontal.
- Would choosing a different point as the origin for the analysis of the data **change** the value of the acceleration?. Explain your answer.
- If the acceleration of an object is zero, does it mean that the velocity is zero?. Explain briefly, by giving an example.
- If the angle of inclination θ=20^0 in the **Figure-(6)**, what is the acceleration (a) of the puck (m) if the puck is released from rest at the top of the inclined plane?. Assume **g**=9.8 m/s^2.